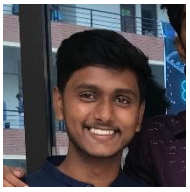
Himanshu Nanda



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Professional Summary

Data Scientist with experience in large-scale fraud detection and alert optimization at AT&T CDO. Specialized in graph-based anomaly detection, experiment design, and agentic AI systems to reduce operational noise and improve investigative efficiency. Strong background in applied machine learning, production systems, and explainable AI.

Technical Skills

- **Programming:** Python, R, PySpark, Java, C++
- **Data & Cloud**: Databricks, Palantir Foundry, MongoDB, MySQL, PostgreSQL, Azure, Docker, Kubernetes
- **Machine Learning:** Deep Learning, Scikit-learn, TensorFlow, Federated Learning, Time-Series Forecasting, XAI
- **Generative & Graph AI:** LangGraph, LangChain, RAG, Vector Search, NetworkX, Community Detection, Anomaly Detection

Educational Qualification

- **B. Tech CSE** : *PES University, Bengaluru* ***(GPA – 8.64/10)*** *2021 – 2025*
- **XIIth** : *City Montessori School, Lucknow* ***(93.25%)*** *June 2020*

Experience

- **AT&T, Bengaluru [CDO – Fraud Prevention Team] – Sr. Associate Data Scientist** *January 2025 - Present*
  - Architected and deployed a **graph-based community intelligence** system to cluster correlated fraud events and suppress redundant alerts, reducing alert volume by 88% while maintaining ~95% detection effectiveness.
  - Designed and optimized a **LangGraph-powered agentic workflow framework** that integrates real-time and historical fraud signals, enabling autonomous triage, contextual investigation, and decision-support automation.
  - Developed intelligent agents that emulate **support engineering workflows to execute context-aware investigative actions**, significantly **reducing manual intervention and operational noise**.
  - Implemented **dynamic threshold optimization using time-series modeling** (ARIMA-based forecasting) to adapt alert sensitivity to evolving fraud behaviors, improving signal stability and reducing false positives.
  - Applied community detection algorithms (**Louvain**) on event graphs and built interactive visualizations (NetworkX, PyViz) to enhance fraud pattern interpretability and investigator prioritization.
  - Conducted structured **Design of Experiments** (DoE) to validate alert strategies and workflow changes, reducing **Mean Time to Identify (MTTI) and Mean Time to Resolve (MTTR).**
  - Contributed to substantial operational cost optimization through intelligent automation and alert reduction strategies, **generating multi-million-dollar savings within a single business unit**.

Projects Undertaken

- **Federated Transfer Learning for Alzheimer Disease Prediction**
  Tools: Python, TensorFlow, Flower, React.js, Node.js, Grad-CAM
  - Developed a **privacy-preserving federated transfer learning framework** for medical imaging, enabling collaborative model training across institutions without centralized data sharing.
  - Bridged the gap between researchers and highly sensitive clinical datasets by ensuring secure, decentralized learning on protected medical data.
  - Integrated **Grad-CAM–based explainable AI** to enhance model interpretability and clinical transparency.
  - Research accepted for publication in an **IEEE-affiliated Springer volume (forthcoming):** https://link.springer.com/book/9789819534975

Certifications

- **Machine Learning Specialization** *July 2023*
- **AWS Educate: Databases** *January 2024*